

Project 1 -Brain-Tasks-Deployment

By Ajmer Singh on March 6th 2026

This project aim to exercise the devOps CI/CD lifecycle. We will explore tools such as docker, ECR registry, AWS EKS, AWS CodeBuild, CodePipeline, Git-hub Version Control, AWS CloudWatch for logs monitoring.

Life Cycle of Project:

GitHub – CodePipeline - CodeBuild – ECR - EKS - CloudWatch Monitoring

Step 1 – To clone the code on local machine

Git clone <https://github.com/Vennilavanguvi/Brain-Tasks-App.git>

```
root@ip-172-31-35-67:/home/ubuntu/ajmer# git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (10/10), done.
remote: Total 11 (delta 0), reused 8 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), 101.89 KiB | 7.84 MiB/s, done.
root@ip-172-31-35-67:/home/ubuntu/ajmer# ls
Brain-Tasks-App
root@ip-172-31-35-67:/home/ubuntu/ajmer# cd Brain-Tasks-App/
root@ip-172-31-35-67:/home/ubuntu/ajmer/Brain-Tasks-App# ls
README.md dist
root@ip-172-31-35-67:/home/ubuntu/ajmer/Brain-Tasks-App#
```

CD to the directory and ls to list the files and folders.

Step 2 – Write a Docker file, as we have source code, so we build an image.

```
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# cat Dockerfile
FROM public.ecr.aws/nginx/nginx:alpine
COPY dist/ /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#
```

FROM pulls the official NGINX image from Docker registry.

COPY moves files from your local project into the container.

EXPOSE tells Docker which port the container will listen on.

CMD default command that runs when the container starts.

Build the docker image

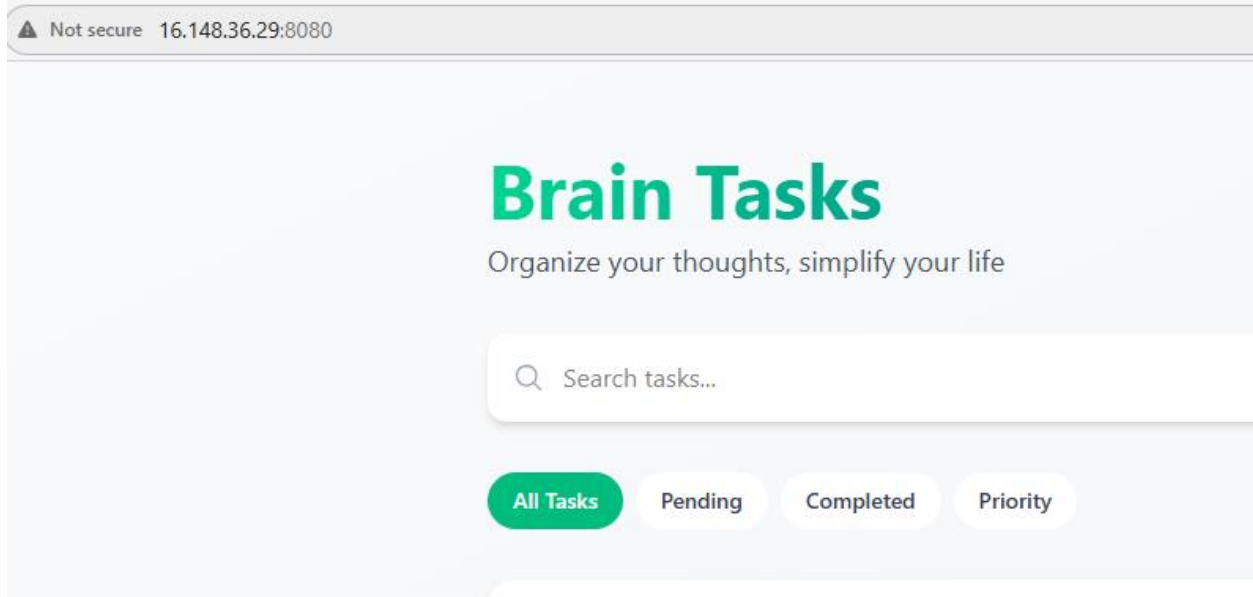
docker build -t mindtrace-aj:v1 .

```

root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# docker run -d -p 8080:80 --name braintest-con 70736e4d65d2
f487f7456aec9b68c3153f65064cf28ca0b7b863af20583c51624918a040b199
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
f487f7456aec   70736e4d65d2                        "/docker-entrypoint..." 9 seconds ago  Up 8 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  braintest-con
11bc0a56dafd   moby/buildkit:buildx-stable-1      "buildkitd --allow-i..." 5 weeks ago   Up 11 hours   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  buildx_buildkit_
e0e96920de03   moby/buildkit:buildx-stable-1      "buildkitd --allow-i..." 6 weeks ago   Up 11 hours   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  buildx_buildkit_
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#

```

Run a test container from the newly build image mindtrace-aj:v1, container name braintest-con, accessed the image by following local ec2 public ip along with port (<http://16.148.36.29:8080/>)



Step 3 – To push the image in AWS ECR.

- Created a repo on ECR named - brain-tasks-app
- List the images <docker images> validate the name of image.
- Tag the image <<docker tag mindtrace-aj:v1 524781515606.dkr.ecr.us-west-2.amazonaws.com/mindtrace-aj:v1>>

This follows aws FEQN followeimg image name.

- docker push 524781515606.dkr.ecr.us-west-2.amazonaws.com/mindtrace-aj:v1

Post tag you would see an addition image on local <<docker images>>

```

root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# docker images
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
524781515606.dkr.ecr.us-west-2.amazonaws.com/brain-tasks-repo:v1 70736e4d65d2                    93MB         26.1MB
jenkins/jenkins:lts                 23b84d1066db                    802MB        288MB        U
mindtrace-aj:v1                     70736e4d65d2                    93MB         26.1MB
moby/buildkit:buildx-stable-1       5601811fde88                    333MB        104MB        U
myacradgrt.azurecr.io/python-app:1.2.0 28091f65421a                    686MB        173MB        U
singajmihub/python-app:1.2.0       28091f65421a                    686MB        173MB        U

```

Where the image ID is the same for both, post push command to test in ECR, check v1 should be available in repo -brain-tasks-repo

brain-tasks-repo

[Summary](#)[Images](#)[Lifecycle policy](#)[Permissions](#)[Repository tags](#)

Images (15) [Info](#)

☐	Image tags	Type	Created at	Image size
☐	-	Image	March 04, 2026, 05:05:55 (UTC+05.5)	0.00
☐	-	Image	March 04, 2026, 05:05:55 (UTC+05.5)	26.08
☐	v1	Image Index	March 04, 2026, 05:05:55 (UTC+05.5)	26.08

Step3 - Created a new project to named <[brain-tasks-repo](#)>

Build projects [Info](#)

[Actions](#)[Create tri](#)

	Name	Source provider	Repository	Latest build status
☐	brain-tasks-repo	AWS CodePipeline	-	Succeeded

Connect the github repo <https://github.com/singh-ajmer-git/MindTrack-aj.git> with in codebuild and supply the buildspec.yml which contains the information of all the phases that we are going to use here for pipeline deployment.

The file buildspec.yml is avilable on git repo which is public

<https://github.com/singh-ajmer-git/MindTrack-aj.git>

Step 4 – Created a repo on Github and pushed all the local files to the repo

git init

git remote -v

git remote add origin <https://github.com/singh-ajmer-git/MindTrack-aj.git>

```
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# git remote -v
origin https://github.com/singh-ajmer-git/MindTrack-aj.git (fetch)
origin https://github.com/singh-ajmer-git/MindTrack-aj.git (push)
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#
```

git add .

git comm -m “ brain task project code adding to the git”

git branch

git checkout dev

git switch dev

git push origin dev

Upon prompt of user id and password, supply the git user id and PAT token from git.

Step 5 – Deploy AWS EKS Cluster with 2 nodes

Clusters (1) [Info](#)

Filter clusters

Cluster name	Status	Kubernetes version	Support period	Upgrade policy	Created
brain-cluster	Active	1.34 Upgrade now	Standard support until December 2, 2026	Extended support	2 hours ago

brain-cluster [Delete cluster](#) [Upgr](#)

End of standard support for Kubernetes version 1.34 is December 2, 2026.

Cluster info [Info](#)

Status Active	Kubernetes version Info 1.34	Support period Standard support until December 2, 2026	Provider EKS
Cluster health 0	Upgrade insights 5	Node health issues 0	Capability issues 0

Overview Resources Compute Networking Add-ons 1 Capabilities Access Observability Update history & backups Tags

Role - eksctl-brain-cluster-cluster-ServiceRole, policy that is must added
“AmazonEKSClusterPolicy”

Imp - configuring kubectl to communicate with the Kubernetes API server of your Amazon Elastic Kubernetes Service cluster.

aws eks --region us-west-2 update-kubeconfig --name brain-cluster

```

root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get ns
NAME                STATUS    AGE
amazon-cloudwatch    Active    80m
default              Active    107m
kube-node-lease      Active    107m
kube-public          Active    107m
kube-system          Active    107m
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get node
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-52-107.us-west-2.compute.internal Ready    <none>    103m   v1.34.4-eks-efcacff
ip-192-168-91-15.us-west-2.compute.internal Ready    <none>    103m   v1.34.4-eks-efcacff
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#

```

Added the CloudWatch monitoring for the cluster, as enabled control plane logging.

Additionally, installed the Fluent Bit DaemonSet that collects pod logs + node logs.

```

curl https://raw.githubusercontent.com/aws-samples/amazon-cloudwatch-container-insights/latest/k8s-deployment-manifest-templates/deployment-mode/daemonset/container-insights-monitoring/quickstart/cwagent-fluent-bit-quickstart.yaml | sed "s/{{cluster_name}}/brain-cluster;/s/{{region_name}}/us-west-2/" |
kubectl apply -f -

```

it deploy the aws supported standard image for FluentBit DaemonSet.

```

root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get node
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-52-107.us-west-2.compute.internal Ready    <none>    103m   v1.34.4-eks-efcacff
ip-192-168-91-15.us-west-2.compute.internal Ready    <none>    103m   v1.34.4-eks-efcacff
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get pods -n kube-system | grep -i cloud
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get pods -n kube-system | grep -i fluent
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get daemonset -n kube-system
NAME                DESIRED    CURRENT    READY    UP-TO-DATE    AVAILABLE    NODE SELECTOR    AGE
aws-node            2          2          2        2             2           <none>           107m
kube-proxy          2          2          2        2             2           <none>           107m
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get pods -n amazon-cloudwatch
NAME                READY    STATUS
cloudwatch-agent-qq456 1/1      Running
cloudwatch-agent-tdc79 1/1      Running
fluent-bit-dwdvf      0/1      CreateContainerConfigError
fluent-bit-mfwbt      0/1      CreateContainerConfigError
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#

```

Verify the Cluste logs in CloudWatch

Log groups
Data sources - new
Summary

▶ Get-set-go with unified data store

Log groups (8)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search
☒ Exact

☐	Log group	Log class	Anomaly d...
☐	/aws/codebuild/brain-tasks-repo	Standard	Configure
☐	/aws/codebuild/brain-task-aj	Standard	Configure
☐	/aws/codebuild/code-pipeline	Standard	Configure
☐	/aws/codebuild/mindbuild	Standard	Configure
☐	/aws/codebuild/singajm-buvi	Standard	Configure
☐	/aws/eks/brain-cluster/cluster	Standard	Configure
☐	/aws/lambda/devopsLambdafunction	Standard	Configure
☐	/aws/lambda/resizetheobject	Standard	Configure

Wrote the deployment.yml & service.yml to deploy and access the deployment. Source code available in git repo -<https://github.com/singh-ajmer-git/MindTrack-aj.git>

Run the test container to validate the cluster functionality.

```
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
brain-app     2/2     2             2           60m
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get pod
NAME          READY   STATUS    RESTARTS   AGE
brain-app-5d7798b8d9-d6vzl  1/1     Running   0           60m
brain-app-5d7798b8d9-jvqt5  1/1     Running   0           60m
nginx-test    1/1     Running   0           90m
test-nginx    1/1     Running   0           80m
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#
```

Step 6 -CodePipeline,

Created a pipeline with the name “brain-tasks-pipeline” which is triggered based on changes in git-hub repo <https://github.com/singh-ajmer-git/MindTrack-aj.git> any push to the repo would trigger this pipeline.

Pipelines Info				 View history
Name	Latest execution status	Latest source revisions	Latest execution star	
 brain-tasks-pipeline	 Succeeded	Source – 50e819c3  test12	1 hour ago	

Added a fresh update to the git repo, pushed it

```
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# git add .
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# git commit -m "final test on brain test app deployment"
[dev 3fa6bc8] final test on brain test app deployment
Committer: root <root@ip-172-31-35-67.us-west-2.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 1 insertion(+)
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# git remote -v
origin https://github.com/singh-ajmer-git/MindTrack-aj.git (fetch)
origin https://github.com/singh-ajmer-git/MindTrack-aj.git (push)
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# git push origin dev
Username for 'https://github.com': singh-ajmer-git
Password for 'https://singh-ajmer-git@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 348 bytes | 348.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/singh-ajmer-git/MindTrack-aj.git
  50e819c..3fa6bc8 dev -> dev
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#
```

Artifacts, the last push to the git was name “test12”, the pipeline get triggered and successfully fetch the information from git, CodeBuild build the image based on instruction that we wrote in buildspec.yml, finally, it deploys the new image as per deployment.yml.

The Pipeline triggered

brain-tasks-pipeline

Pipeline

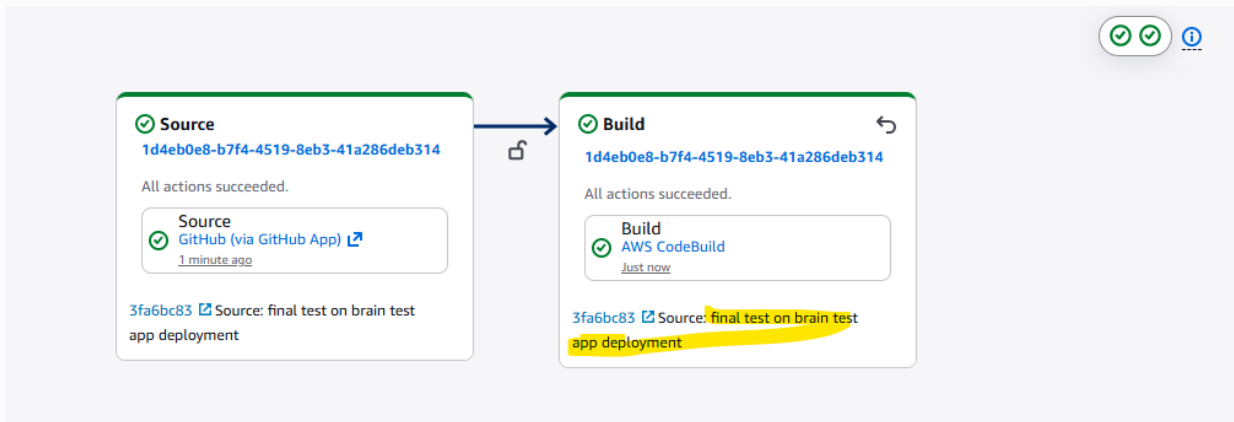
Executions

Triggers

Settings

Tags

Stage



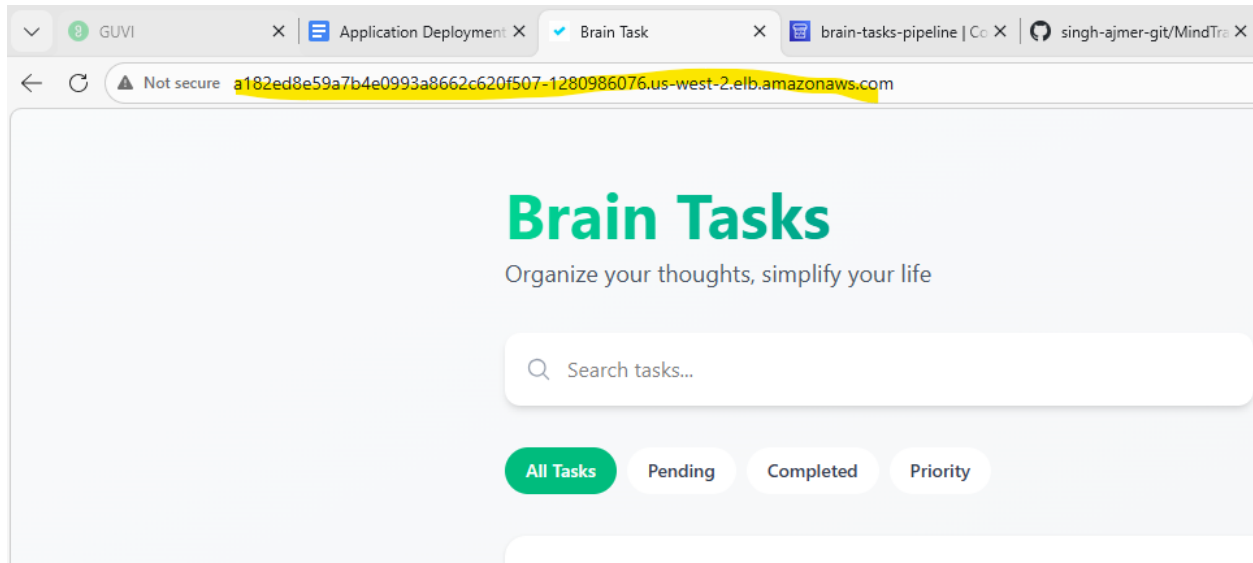
Pods running

```
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
brain-app     2/2     2             2           87m
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
brain-app-5bcb8d844f-nzzkl         1/1     Running   0           2m30s
brain-app-5bcb8d844f-ptp4q         1/1     Running   0           2m28s
nginx-test                          1/1     Running   0           117m
test-nginx                         1/1     Running   0           106m
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#
```

SVC confirmed

```
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App# kubectl get svc
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
brain-app-service  LoadBalancer  10.100.172.124  a182ed8e59a7b4e0993a8662c620f507-1280986076.us-west-2.elb.amazonaws.com  80:30632/TCP    3m
kubernetes     ClusterIP     10.100.0.1    <none>         443/TCP          147m
root@ip-172-31-35-67:/home/ubuntu/Brain-Tasks-App#
```

Accessible via external address of load balancer



In this project we build the image by using docker build with the help of Dockerfile. Testing by running a local container and exposing it to a local host. Pushed the image manually to AWS ECR. Create a project in CodeBuild to facilitate the infra to build via CodePipeline. Deploy the cluster along with CloudWatch. Finally, prepare a CodePipeline based on trigger from git and testing the outcomes by accessing the web via cluster load balancer.